

TAHMID ZAMAN TAHI

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Research focus: Ph.D. researcher developing physics-guided machine learning and statistical signal processing for wireless communications, sensing, and localization, with current work in near-field RIS systems, low-SNR signal intelligence, and reliable V2X.

Education

Virginia Commonwealth University, Richmond, VA
Ph.D. in Electrical Engineering; Advisor: Dr. Yanxiao Zhao

Expected Dec 2028
GPA: 4.0/4.0

The George Washington University, Washington, DC
M.S. in Electrical Engineering

May 2024
GPA: 3.7/4.0

Islamic University of Technology, Gazipur, Bangladesh
B.Sc. in Electrical and Electronic Engineering

Mar 2021

Research Experience

Graduate Research Assistant

Virginia Commonwealth University, Department of Electrical and Computer Engineering

Jan 2025 – Present
Richmond, VA

Collision-Aware Near-Field RIS Localization

Ongoing Ph.D. research

- Formulating multi-source 3D localization under limited sounding, unknown source count, hardware impairments, and same-pilot collisions in near-field RIS-assisted systems.
- Developing a hybrid learned-physics estimator using pilot-conditioned matched-filter/GLRT maps, activity and multiplicity inference, candidate-set ranking, angle-range refinement, joint gain refitting, and task-aware RIS schedule selection.
- Building bank-disjoint PyTorch/Sionna/Slurm simulation and evaluation pipelines and implementing classical and learned baselines for source-count, strict-3D accuracy, collision robustness, and latency comparisons.

BR-FiLM for Low-SNR Automatic Modulation Recognition

MILCOM 2026 (under review)

- Proposed a modular bounded-residual FiLM block that applies gated channel-quality corrections at convolutional, recurrent, and dense stages while preserving the original I/Q-driven representation path.
- Improved MCLDNN from 61.79% to 67.74% overall accuracy and from 37.12% to 46.44% for $\text{SNR} \leq 0$ dB on 44,000 RadioML 2016.10a test examples; paired statistical tests confirmed the gains.
- Evaluated accuracy, per-class behavior, conditioning gates, and deployment cost against convolutional, recurrent, and transformer-style baselines; added 50,955 parameters and 0.65 ms batch-1 GPU latency.

Context-Aware Functional Security for V2X Systems

IEEE ITEC+EATS 2026

- Led development of a context-conditioned monitoring framework combining domain-informed KPI features, safety-weighted reconstruction scoring, and per-context threshold calibration for changing V2X operating regimes.
- Designed synthetic stress tests for jamming, compute DoS, spoofing, and control-plane misconfiguration and benchmarked six unsupervised model families under 5-fold cross-validation.
- Found that Slice $\times$ Weather calibration improved true-positive rate by roughly 2–4 $\times$ at a fixed 1% false-positive rate across all models, making calibration more consequential than model complexity in the evaluated regime.

Energy-Efficient CSI Estimation and RIS Phase Prediction

ICCCN 2026 (accepted)

- Contributed to a DSRNetV2-PhaseNet framework that reconstructs 1,024-element RIS channel state information from 64 active elements and predicts RIS phase shifts for large-scale NOMA systems.
- Evaluated a progressive multi-objective training strategy that transitions from channel-estimation accuracy to rate-aware optimization while retaining a fairness constraint for the weaker user.
- Achieved 12.79 bps/Hz sum rate, 2.46 Mbits/J energy efficiency, and 99.95% user fairness; improved throughput by 32% and energy efficiency by 37% over fixed-weight training.

Selected Publications and Manuscripts

- T. Z. Tahi**, S. S. Alam, H. Tang, Y. Zhao, J. Huang, and M. Song, “BR-FiLM: Bounded Residual Channel-Quality Conditioning for Automatic Modulation Recognition,” *under review at the IEEE Military Communications Conference (MILCOM)*, 2026.
- S. S. Alam, Y. Zhao, H. Tang, **T. Tahi**, W. Wang, N. K. Dhar, and M. Song, “Deep Learning-Enabled Energy-Efficient Joint CSI Estimation and Phase Shift Prediction in Large-Scale RIS-Assisted NOMA Systems,” *accepted for presentation at the 35th International Conference on Computer Communications and Networks (ICCCN)*, 2026.
- T. Z. Tahi**, S. Saha, M. Zaman, Y. Zhao, and S. Abdelwahed, “Context-Aware Functional Security for V2X-Enabled Connected Vehicles,” in *Proc. IEEE Transportation Electrification Conference & Expo and Electric Aircraft Technologies Symposium (ITEC+EATS)*, 2026.
- N. K. Surendran, S. S. Alam, H. Tang, **T. Tahi**, W. Wang, and Y. Zhao, “Enhancing IRS Localization via Deep Learning-Based AOA and Distance Estimation,” *Journal of AI-Driven Communication Engineering*, vol. 1, pp. 18–28, 2025.
- T. Z. Tahi**, “Resource Allocation in C-V2X: A Review,” arXiv:2401.15756, 2024.

Selected Academic Projects

IoT-Enabled Smart Home HVAC Control System 2025

- Co-developed an end-to-end IoT HVAC controller using an Arduino MKR WiFi 1010, DHT11 and UV sensors, ThingSpeak, and cloud-hosted MATLAB logic to generate AC, heater, dehumidifier, and blind states.
- Implemented periodic sensing, Wi-Fi recovery, cloud data ingestion, visualization, state-change detection, and email alerts; validated all control branches using controlled temperature, humidity, and lighting stimuli.

Learning-Based Path Planning for Autonomous Systems 2024

- Built dynamic navigation environments and trained a deep Q-network agent for goal-directed path planning under changing conditions; compared the learned policy quantitatively with classical planning algorithms.

Additional Experience

Engineer, Network Infrastructure Nov 2021 – Dec 2022 *ADN Telecom Limited* Dhaka, Bangladesh

- Engineered RF link budgets, SNR analyses, feasibility studies, troubleshooting, and technical documentation for enterprise network deployments across Bangladesh.
- Co-led deployment and user-acceptance testing of an IoT network-management system monitoring power devices at 41 universities, delivering the project two months ahead of schedule.

Graduate Teaching Assistant Jan 2023 – May 2024 *The George Washington University, Department of Electrical and Computer Engineering* Washington, DC

- Led recitations and office hours for Circuits, Signals, and Systems and supported Digital Signal Processing, Engineering Analysis, and Computer Organization for approximately 60 students per semester.

Technical Skills

Programming: Python, MATLAB, C, Bash

Machine Learning and Scientific Computing: PyTorch, TensorFlow/Keras, NumPy, SciPy, scikit-learn, Pandas, Matplotlib

Wireless and Signal Processing: Sionna, Simulink, channel modeling, array and subspace processing, OFDM, RIS, near-field localization, automatic modulation recognition, V2X/C-V2X

Research Systems: Linux, Git, Slurm, CUDA, JupyterLab, Ray, Optuna, LaTeX

Service, Honors, and Leadership

Reviewer, IEEE Military Communications Conference (MILCOM) 2026

Graduate Fellowship Award, The George Washington University 2023

Chief Operations Officer, IUT Career and Business Society 2018–2021

Mentored 100+ students and helped organize a national business competition.